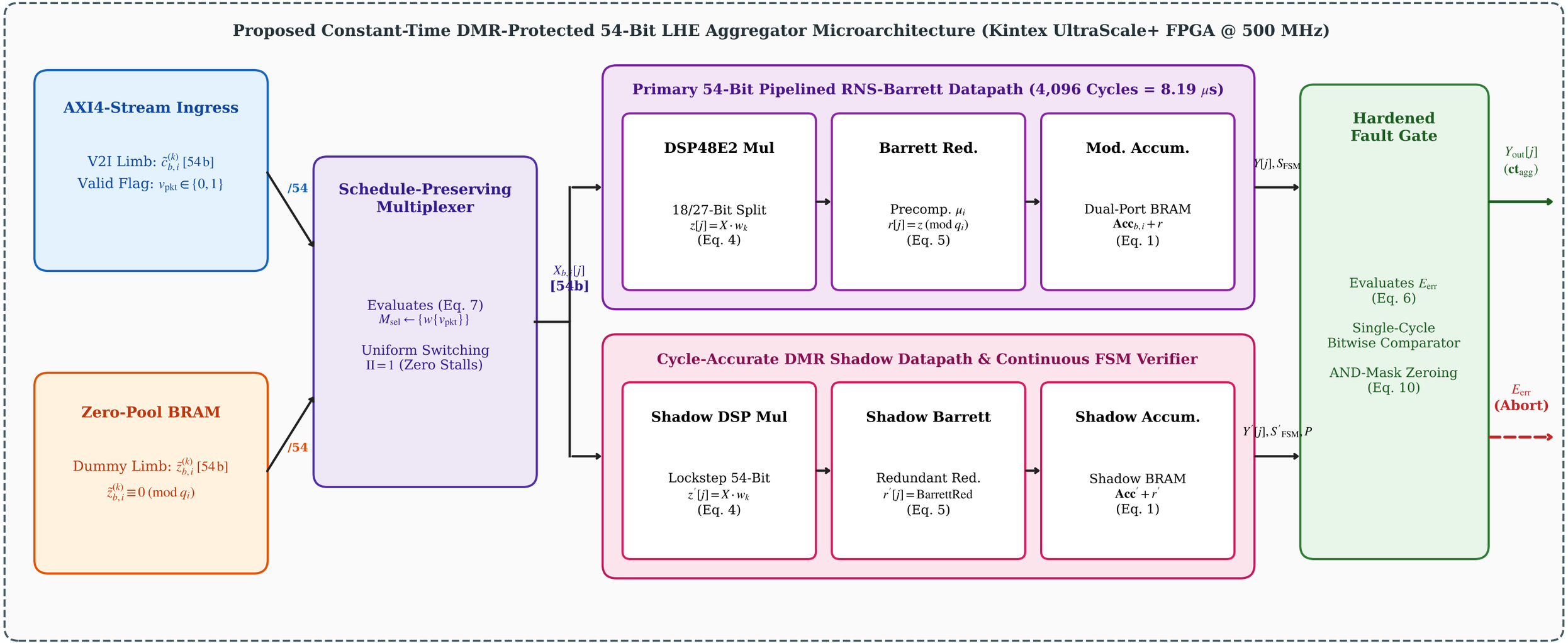

Proposed Constant-Time DMR-Protected 54-Bit LHE Aggregator Microarchitecture (Kintex UltraScale+ FPGA @ 500 MHz)
AXI4-Stream Ingress
V2I Limb: $\tilde{c}_{b,i}^{(k)}$ [54b]
Valid Flag: $v_{\text{pkt}} \in \{0,1\}$
/54
Zero-Pool BRAM
Dummy Limb: $\tilde{z}_{b,i}^{(k)}$ [54b]
$\tilde{z}_{b,i}^{(k)} \equiv 0 \pmod{q_i}$
/54
Schedule-Preserving Multiplexer
Evaluates (Eq. 7)
$M_{\text{sel}} \leftarrow \{w\{v_{\text{pkt}}\}\}$
Uniform Switching
II = 1 (Zero Stalls)
$X_{b,i}[j]$
[54b]
Primary 54-Bit Pipelined RNS-Barrett Datapath (4,096 Cycles = 8.19 μs)
DSP48E2 Mul
18/27-Bit Split
$z[j] = X \cdot w_k$
(Eq. 4)
Barrett Red.
Precomp. μ_i
$r[j] = z \pmod{q_i}$
(Eq. 5)
Mod. Accum.
Dual-Port BRAM
$\mathbf{Acc}_{b,i} + r$
(Eq. 1)
Cycle-Accurate DMR Shadow Datapath & Continuous FSM Verifier
Shadow DSP Mul
Lockstep 54-Bit
$z'[j] = X \cdot w_k$
(Eq. 4)
Shadow Barrett
Redundant Red.
$r'[j] = \text{BarrettRed}$
(Eq. 5)
Shadow Accum.
Shadow BRAM
$\mathbf{Acc}' + r'$
(Eq. 1)
$Y[j], S_{\text{FSM}}$
$Y'[j], S'_{\text{FSM}}, P$
Hardened Fault Gate
Evaluates E_{err}
(Eq. 6)
Single-Cycle
Bitwise Comparator
AND-Mask Zeroing
(Eq. 10)
$Y_{\text{out}}[j]$
($\mathbf{ct}_{\text{agg}}$)
E_{err}
(Abort)